

Rising Waters

Flood Management System — AI/ML Project

Brainstorming & Idea Prioritization

Phase 1: Brainstorming & Ideation

IBM Cloud

XGBoost 96.55%

Skill Wallet

AI/ML Track

1. Problem Domain

Flooding is one of the most devastating natural disasters globally, impacting millions of lives every year. The absence of timely and accurate early-warning systems significantly amplifies the damage caused by floods. Traditional forecasting methods lack real-time intelligence and predictive accuracy.

2. Brainstorming Ideas Generated

The team conducted an ideation session using the Design Thinking framework. The following ideas were generated:

- Real-time flood sensor network with IoT integration
- Satellite imagery analysis using computer vision
- **ML-based flood prediction using meteorological data (SELECTED)**
- Social media monitoring for early community alerts
- Hydrological model simulation platform
- Mobile app with location-based flood risk alerts

3. Idea Prioritization Matrix

4. Selected Idea

Machine Learning-Powered Flood Prediction System — Using meteorological features (annual rainfall, seasonal rainfall, cloud visibility, humidity, temperature, and river level) to classify flood vs. no-flood events using XGBoost. Deployed as a Flask web application on IBM Cloud.